

# Applications of Microbiology with Biostatistics

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**Science**

**May, 2026**



# Applications of Microbiology with Biostatistics (A27781)

**Short Title:** Applic Microbiol with Biostats  
**Faculty** Science and Computing  
**Department** Science  
**Cluster** Biology  
**Author** LCOFFEY  
**Credits** 5

**Level:** Advanced

## Description of Module / Aims

This module exposes the student to microbiological and molecular methods and their applications in a broad range of areas such as food, pharmaceutical and environmental. It also introduces the student to the use of microbes as analytical tools, and a range of topical and emerging microbiological issues and applications.

## Programmes

|           |                                                                                |           |
|-----------|--------------------------------------------------------------------------------|-----------|
| BIOL-0027 | BSc (Hons) in Molecular Biology with Biopharmaceutical Science<br>(WD_SMBIO_B) | 4 / 7 / M |
|-----------|--------------------------------------------------------------------------------|-----------|

## Indicative Content

- Review of culture-based microbiological methods for analysis of pharmaceuticals, foods, and environmental samples: use in applications such as sterility testing, bioburden, environmental monitoring, hygiene testing etc
- Nucleic acid-based methods for detection, identification and enumeration of micro-organisms, such as 16S rRNA gene sequencing, real-time PCR, PFGE, whole genome sequencing etc.
- Use of micro-organisms as analytical tools in applications such as clinical sample analysis, toxicity testing, MIC determination etc.
- Topical issues in microbiology, e.g. emerging diseases, drug discovery, the Human Microbiome Project etc.
- Topical epidemics and the human response e.g. Ebola, Zika
- Statistical interpretation of microbiological data

## Learning Outcomes

*On successful completion of this module, a student will be able to:*

1. Compare and contrast traditional and more modern microbiological methods.
2. Evaluate a wide range of applications of microorganisms and their products.
3. Evaluate appropriate analytical methods for microbiological analysis of specific samples.
4. Comment on new microbiological techniques and their future potential in industry.
5. Interpret statistical analysis of microbiology data.
6. Compare emerging pathogens and other topical microbiological issues.

## Learning and Teaching Methods

- Lectures.
- Laboratory practicals
- Group discussions.
- Self-directed study.
- Formative assessment.
- Peer-learning.

## Assessment Methods

|                           | Weighing | Outcomes Assessed |
|---------------------------|----------|-------------------|
| Final Written Examination | 60%      | 1,4,5,6           |
| Continuous Assessment     | 40%      |                   |
| <i>Lab Report</i>         | 40%      | 2,3               |

## Assessment Criteria

**<40%:** No understanding of applications of microbiology or biostatistics. Unable to carry out critical thinking.

**40%-49%:** Demonstrate a limited knowledge of microbiology applications and biostatistics. May have some difficulty with critical thinking.

**50%-59%:** A good understanding of analytical microbiology and topical and emerging microbiological issues with demonstration of some critical thinking.

**60%-69%:** In addition to the above, good critical thinking skills and showing an ability to interpret statistical analysis of microbiology data.

**70%-100%:** A highly developed and extensive understanding of analytical microbiology and topical and emerging microbiological issues.

**Note:** In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

## Learning Modes

| Learning Mode        | F/T Hours | P/T Hours |
|----------------------|-----------|-----------|
| Lecture              | 36        |           |
| Practical            | 24        |           |
| Independent Learning | 75        |           |

## Essential Material

- "Australian government." Guidelines for sterility testing. <http://www.tga.gov.au/docs/pdf/sterilit.pdf>
- "Moodle." [www.moodle.wit.ie](http://www.moodle.wit.ie)
- Armstrong, R.A. and A.C. Hilton. \_Statistical Analysis in Microbiology: StatNotes\_. USA: Wiley-Blackwell, 2014.
- Collins, C.H. \_Collins and Lyne's microbiological methods\_. 8th Edition. London: Arnold, 2003.
- Madigan, M.T. \_Brock Biology of Microorganisms\_. 14th Edition. USA: Pearsen , 2014.
- Zar, J.H. \_Biostatistical Analysis\_. 5th ed.. USA: Prentice Hall, 2007.

## Supplementary Material

- Bisen, S.P. \_Microbes: concepts and applications\_. USA: Wiley-Blackwell, 2012.

## Requested Resources

- Lecture Room: Loose Seated
- Room Type: Biology Lab